

Bounded Collatz Correction and Reciprocal Orbit Sums

Collatz proof project*

September 5, 2026

Abstract

For a positive integer shortcut Collatz orbit, boundedness of the normalized affine correction is equivalent to convergence of the sum of reciprocal orbit values. Two finite inequalities prove the equivalence: a telescoping estimate that charges reciprocal mass to correction increments, and an exponential upper bound on correction in terms of accumulated reciprocal mass. No strict odd-step density gap is assumed. Bounded correction consequently forces escape from every finite set. The equivalence characterizes a class of hypothetical counterexamples; it neither excludes that class nor shows that every escaping orbit belongs to it.

1 Definitions and exact characterization

Let $N \in \mathbb{N}$ be positive and define

$$T(x) = \begin{cases} x/2 & x \text{ even,} \\ (3x+1)/2 & x \text{ odd,} \end{cases} \quad n_t = T^t(N).$$

All n_t are positive integers. Write j_t for the number of odd terms among $n_0, \dots, n_{t-1}$, and let d_t be the integer affine correction determined by

$$2^t n_t = 3^{j_t} N + d_t.$$

Set

$$c_t = \frac{d_t}{3^{j_t}}, \quad r_t = \frac{1}{n_t}, \quad S_L = \sum_{t=0}^{L-1} r_t.$$

Thus $c_0 = 0$, $c_t \geq 0$, and c_t is nondecreasing. The existing Lean predicate **Supercritical N** means precisely $\sup_t c_t < \infty$. Its name does not add any density assumption.

Theorem 1. *For every positive integer N ,*

$$\sup_t c_t < \infty \iff \sum_{t=0}^{\infty} \frac{1}{n_t} < \infty.$$

The proof below uses finite estimates valid for every positive seed. In particular, neither direction assumes that the orbit diverges, has a limiting odd-step density, or satisfies a uniform density gap.

*Prepared with Codex (OpenAI). All central statements are formalized in Lean. No claim of literature priority is made.

2 Finite inequalities

The affine identity gives

$$\frac{2^t}{3^{j_t}} = (N + c_t)r_t. \quad (1)$$

The correction cocycle gives

$$c_{t+1} - c_t = \begin{cases} 0 & n_t \text{ even,} \\ (N + c_t)r_t/3 & n_t \text{ odd.} \end{cases} \quad (2)$$

These identities hold in $\mathbb{R}$; the divisions are legitimate because $n_t > 0$.

Lemma 2. *For every $L \in \mathbb{N}$,*

$$NS_L \leq 6c_L + N(r_L - r_0). \quad (3)$$

Proof. We prove the one-step estimate

$$Nr_t \leq 6(c_{t+1} - c_t) + N(r_{t+1} - r_t).$$

If n_t is even, then $r_{t+1} = 2r_t$ and the correction increment vanishes, so equality holds. If n_t is odd, equation (2) and $c_t \geq 0$ give

$$6(c_{t+1} - c_t) \geq 2Nr_t.$$

Adding $N(r_{t+1} - r_t)$ gives at least Nr_t , because $r_{t+1} \geq 0$. Sum over $0 \leq t < L$ and telescope, using $c_0 = 0$. $\square$

The constant six is a convenient uniform choice; optimality is not asserted.

Lemma 3. *For every $L \in \mathbb{N}$,*

$$N + c_L \leq N \exp(S_L). \quad (4)$$

Proof. In both parity cases equation (2) implies

$$N + c_{t+1} \leq (N + c_t)(1 + r_t) \leq (N + c_t)e^{r_t}.$$

All factors are nonnegative. Induction from $c_0 = 0$, using $e^{a+b} = e^a e^b$, proves the claim. The Lean proof uses the existing one-step bound `idealC 1 n_t ≤ 1` and the cocycle; the sharper odd-step constant $1/3$ is unnecessary here. $\square$

3 Infinite sums and escape

Proof of Theorem 1. Suppose first that $c_t \leq B$ for every t . Positivity and integrality give $0 \leq r_t \leq 1$. Lemma 2 therefore implies

$$S_L \leq \frac{6B}{N} + 1 \quad \text{for every } L.$$

The nonnegative partial sums are bounded, so the series converges. Lean uses mathlib's `summable_of_sum_range_le`.

Conversely, suppose $S = \sum_{t \geq 0} r_t < \infty$. Nonnegativity gives $S_L \leq S$. Lemma 3 yields

$$N + c_L \leq Ne^S \quad \text{for every } L,$$

which bounds c_L . The formal proof uses the upper bound Ne^S for c_L ; the displayed estimate also gives $c_L \leq N(e^S - 1)$. $\square$

Corollary 4. *If $N > 0$ and $\sup_t c_t < \infty$, then $n_t \rightarrow +\infty$. More precisely, for every $B \in \mathbb{N}$ there is a time after which $n_t \geq B$ always holds.*

Proof. The terms of a convergent series tend to zero, so $r_t \rightarrow 0$. Eventually $r_t < 1/(B+1)$, which forces $n_t > B$. The Lean conclusion uses the equivalent non-strict at-top formulation. $\square$

In particular, a positive orbit that is eventually periodic cannot have bounded correction. This last sentence follows immediately from Corollary 4; it is not a separately named declaration in the new module.

4 What this changes in the proof program

The earlier geometric inverse-drift bound remains a sufficient condition for bounded correction. Theorem 1 supplies an exact orbit-value condition instead, and does not require exponential growth. For example, an independently proved envelope $n_t \geq A(t+1)^2$ with $A > 0$ would suffice by comparison with a convergent square reciprocal series. This comparison is an explanatory consequence, not a separate formal theorem in this module, and no such envelope is claimed for any natural Collatz seed.

The main unresolved question is now visible in orbit terms: can an escaping positive integer Collatz orbit have $\sum_t 1/n_t = \infty$? Mere escape does not imply reciprocal summability for general positive sequences. Even settling this coverage question would leave the task of excluding bounded-correction orbits, as well as the separate nontrivial-cycle problem. Thus the present result is a characterization, not a proof of Collatz or an exclusion theorem.

5 Lean declarations and reproducibility

The implementation is `lean/Collatz/Reciprocal.lean`. It imports the project’s ideal-correction bounds and `mathlib`’s real exponential and nonnegative-series facts. The primary theorem is:

```
supercritical_iff_summable_reciprocal
For  $N > 0$ , Supercritical N if and only if
Summable (orbitReciprocal N).
```

Here `orbitReciprocal N t` is $1/(T^t(N) : \mathbb{R})$.

Result	Declaration in namespace <code>Collatz</code>
Equation (1)	<code>inverse_drift_eq_reciprocal</code>
Lemma 2	<code>reciprocal_sum_ledger</code>
Lemma 3	<code>idealC_exp_reciprocal_bound</code>
Theorem 1	<code>supercritical_iff_summable_reciprocal</code>
Corollary 4	<code>supercritical_orbit_tendsto_atTop</code>

The module also contains each implication separately and the local one-step ledger. No `sorry`, new axiom, or `native_decide` is used.

From the repository root, rebuild and audit with:

```
cd lean
lake build Collatz.Reciprocal
cd ..
python3 analysis/audit_theorems.py --build
```

The pinned Lean toolchain is `leanprover/lean4:v4.27.0-rc1`. The selected audit now contains 48 declarations, including the two finite estimates, the equivalence, and the escape corollary. Their reported dependencies use only `propext`, `Classical.choice`, and `Quot.sound`. The JSON audit records exact printed types and repository provenance. It checks selected modules and their Lean-reported dependencies; it is not an independent kernel replay or a rebuild of every nested library.

6 Sources and scope of attribution

The definitions and affine cocycle are in the existing project module `Collatz.Ideal`; the previously established geometric criterion is in `Collatz.IdealBounds`. Nonnegative-series and exponential lemmas are supplied by the pinned `mathlib` checkout. All new mathematical arguments used here are displayed above and implemented in `Collatz.Reciprocal`. No external literature-priority claim is made.